

C Programming Exam: Simple Banking System

Total Points: 20

Duration: 90 minutes

□ Instructions

- Use **standard C libraries** only.
- Store data in memory only (no file I/O).
- Use **pointers** appropriately where required.
- Validate inputs and handle edge cases.
- Write modular, clean, and well-commented code.

Question 1: Define Account Structure and Main Menu (3 Points)

Task:

- Define an **Account** structure with:
 1. **accountNumber** (int)
 2. **holderName** (char[50])
 3. **balance** (float)
- Show a menu with options:
 1. Create Account
 2. Deposit Money
 3. Withdraw Money
 4. Display Accounts
 5. Search Account
 6. Exit

Points Distribution:

- Structure definition: 1 Point
- Menu loop and structure: 2 Points

Question 2: Create and Display Accounts (4 Points)

Task:

- Use an array to store up to 5 accounts.
- Validate:
 - Unique account number
 - Non-negative initial balance
- Display all accounts in table format.

Points Distribution:

- Add account with validation: 2 Points
- Display all accounts: 2 Points

Question 3: Deposit and Withdraw Operations (4 Points)

Task:

- Use **pointers** to directly modify the account's balance.
- Validate account number and positive amounts.

Deposit:

- Prompt for account number and deposit amount.
- Use a pointer to update balance.

Withdraw:

- Prompt for account number and amount.
- Ensure sufficient balance using pointer logic.

Points Distribution:

- Pointer logic to update balance (2 Points)
 - Validation and correct updates (2 Points)
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Question 4: Pointer-based Account Search (4 Points)

Task:

- Write a function that **returns a pointer to an account** given an account number.
- Use this pointer in both deposit and withdraw operations.
- Display account details if found, or show “Account not found”.

Points Distribution:

- Correct use of pointer return (e.g., `Account* findAccount(...)`) – 2 Points
- Use returned pointer to display or modify data – 2 Points

Question 5: Input Validation and Menu Handling (3 Points)

Task:

- Prevent adding more than 5 accounts.
- Handle invalid menu choices gracefully.
- Provide meaningful error messages.

Points Distribution:

- Max capacity check: 1 Point
 - Invalid choice handling: 1 Point
 - Friendly messages: 1 Point
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☐ **Submission Instructions**

- Submit `.c` source code file(s).
- Code must compile and run without errors.
- Comment key sections explaining your logic.
- Package all files into a `.zip` named:
`C_BankingPointer_<YourName>.zip`